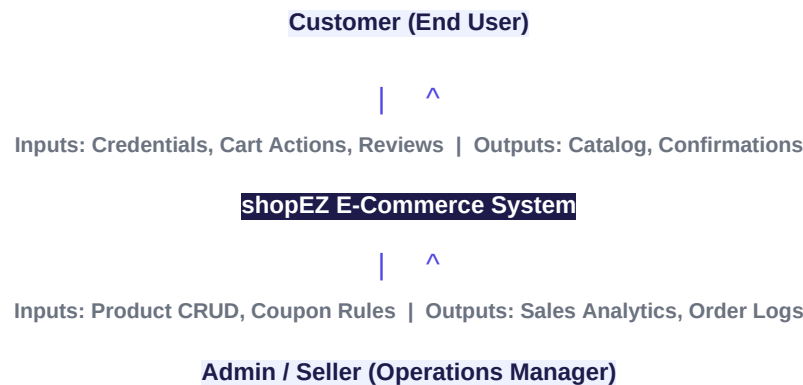


Data Flow Diagrams and User Stories

4. Data Flow Diagrams (DFD)

Level 0 DFD: Context Diagram



Level 1 DFD: Process Breakdown

- User Authentication Flow:**
- Customer inputs email/password -> Express Router -> [authMiddleware.js](#) -> Database User validation -> Server returns JWT token stored in cookie.
- Product Browsing Flow:**
- React catalog triggers GET `/api/products?search=...` -> [productController.js](#) -> Database text search with filters -> Server returns filtered product list.
- Cart Checkout Flow:**
- Customer places order -> React initiates POST `/api/payment/create-intent` -> Server processes mock Stripe transaction -> Server saves Order to [Order.js](#) -> Server decrements Product stock counts in

3. Agile User Story Traceability Matrix

User Goal	Epic Activity Mappings	Mapped USN ID
Register & Log In	Account creation, password encryption, secure session JWT creation	USN-001, USN-002, USN-003
Browse Catalog	Catalog text searches, dynamic autocomplete, categories filtering, ratings sort	USN-004
Manage Cart / Wishlist	Add items, change quantities, save favorites, One-Click Move to Cart	USN-005, USN-006
Purchase Items	Checkout stepper, validate promo coupons, Stripe mock transaction payment	USN-007, USN-008
Track & Review	6-step order tracking progress bar, verified purchase reviews, upvotes	USN-009, USN-010